SPECIAL PROJECT 1

Calculate the Mach number, temperature, pressure variation along the nozzle of your choice starting from the x=0 using the Chapter 8 material of Shapiro. The shape can be approximated using a polynomial. You will use the code sent to generate your own nozzle geometry contour.

Write the code to include all the variables that can be manipulated. Next run the program for the following specific cases, and then compare the results to each other. Three separate plots will be required as the results, one for each of the variations along the x.

a) Assume that the flow is isentropic.
b) Assume that the friction coefficient is f = 0.005, but there is no heat transfer
c) Assume that the heat transfer is not zero, but there is no friction.
d) Assume that there are both the effects of the friction and heat transfer are present.

Hint: Very near the throat the delta-x variation is required to be very small. At Mach=1, the equations become indefinite.

Use Inlet total pressure 50 MPA (gage), total temperature 3000 K

REQUIRED OUTPUT

The output for this project will be the WORD file that contains the programs used and the Figures that show the variation of the

- Mach number
- Temperature and
- Pressure

Along the axis of the nozzle for the 4 different scenarios (a thru d). Example variations are shown for the isentropic case at the bottom of the file.

- Your plots abscissa should be the distance starting at the throat section (x=0).
- Plot the pressure variation using the semilogy command. This will plot the pressure values in a logarithmic scale allowing variations to be easily visible.
- Make sure that the Mach number, temperature, and pressure changes are clearly visible with the values chosen for the friction and the heat addition.
- Make sure to use small variation for the dx increments so that you will have a smooth variation along the axis. Therefore there will not be straight line sections and no glitches in the plots.
- You may want to plot the Mach number, temperature, and pressure variations for the 4 cases (a thru d) on the same plot. This approach will require you to have 3 plots in your file, but this is not a requirement. Otherwise you will have 4 plots for each of the variables (these should be on the same page for easy grading by our TA).